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Combining review elements for modelling various multi-criteria collaborative recommendation models

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Abstract

Traditional single-criterion recommender systems rely on overall ratings, failing to capture accurate user preferences. While multi-criteria recommender systems (MCRSs) address this by leveraging explicit or implicit criteria, existing studies predominantly focus on single review elements, overlooking the potential of combining multiple review elements for richer insights. This paper bridges this gap by proposing novel MCRS models that integrate diverse review elements—such as implicit ratings, aspects, and helpfulness—to enhance recommendation accuracy. A key innovation lies in a novel user profile modelling approach that dynamically combines these elements, enabling granular preference analysis. Comprehensive experiments on the large-scale Amazon dataset demonstrate that the Trust-based Multi-Criteria Similarity with Average Value (TMCSAV) model outperforms all proposed models and the state-of-the-art baselines, achieving the lowest prediction errors (MAE: 0.7473, RMSE: 0.9966) and superior relevance identification (F1-score: 0.65). By prioritising trustworthy users and semantically clustered aspects, TMCSAV mitigates data sparsity and noise, validating the importance of multi-element integration. This work advances MCRS theory through hierarchical aspect clustering and trust-aware paradigms while offering practical value for industries reliant on personalised recommendations, from e-commerce to streaming services.

Keywords: Multi-criteria recommender systems, Implicit criteria, Review elements, User profile modelling, trust-based modelling, Large-scale datasets

Introduction

Recommender Systems (RSs) are pivotal in the digital landscape, aiding users in discovering items that align with their preferences. RSs have been widely adopted across various platforms, including e-commerce, streaming services, and social networks, to provide personalised recommendations. The three primary recommendation approaches are content-based (CB), collaborative filtering (CF), and hybrid systems. Among these, CF remains predominant due to its effectiveness in leveraging user similarity to generate recommendations [1, 2].

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Traditional CF approaches typically rely on a single overall rating provided by users to model preferences. While these systems perform well when sufficient rating data is available, they often fail to capture the underlying reasons behind a user's rating [3]. The overall rating reflects the user's general satisfaction with an item but does not provide insights into the specific aspects that influenced the rating. For example, in the movie domain, some users may prefer a movie for its story and cast, while others may prioritise performance and music. This lack of granularity in single-criterion systems can lead to suboptimal recommendations, as the system cannot distinguish between users who provide the same overall rating but have fundamentally different preferences [4].

Multi-criteria recommender systems (MCRSs) have been introduced to address existing limitations. These systems enhance recommendation accuracy and personalisation by leveraging explicit or implicit criteria to capture a comprehensive view of user preferences [2, 5, 6]. Explicit criteria, which are predefined and easily analysed, provide a clear structure. However, they may limit the expression of full user preferences. Conversely, implicit criteria, derived from unstructured user-generated reviews through natural language processing (NLP), offer a richer, user-driven insight into preferences, capturing subtleties that explicit criteria might miss [7]. Although studies like those [3, 8–10] have demonstrated the effectiveness of explicit criteria, the dynamic nature of implicit criteria (i.e., also called elements) from online reviews—such as *total review polarity scores, aspects, emoticons* and *contexts*—presents a valuable opportunity to refine further and enhance the quality of recommendations.

Recent studies have demonstrated the effectiveness of implicit ratings in MCRSs for improving recommendation accuracy [11–13]. Despite these advancements, most existing research focuses on employing a single review element, such as *implicit ratings or aspects*. For instance, the works of [11, 14, 15] generate implicit ratings from reviews to enhance RSs performance, while [13, 16–18] employ the *aspects* review element. This leaves a significant gap in exploring the potential of combining multiple review elements to create more robust and accurate MCRS models. This study addresses this gap by proposing various MCRS models that integrate multiple review elements, such as *implicit ratings*, *aspects*, and *helpfulness*, to enhance the recommendation process. The primary research questions guiding this study are as follows.

- RQ1: How can multiple review elements be effectively integrated into user profiles to model user preferences in MCRSs?
- RQ2: What is the impact of combining implicit criteria on the predictive accuracy of MCRS models?
- RQ3: How do different combinations of review elements influence the performance of MCRSs in terms of prediction accuracy?

This study contributes to the field of MCRSs in several significant ways:

- Novel user profile modelling—a new approach for modelling user profiles by integrating multiple review elements.
- Multiple MCRS models—developing various MCRS models based on different combinations of review elements to evaluate their impact on recommendation accuracy.

- Comprehensive evaluation—we conducted extensive experiments on the large-scale Amazon dataset to rigorously evaluate the MCRS models' performance. Our evaluation framework includes comparisons against state-of-the-art baseline models to identify the most effective approach.

The rest of this paper is organised as follows. Section 2 provides background on multi-criteria RSs and review elements. Section 3 reviews related work in the field. Section 4 describes the methodology for developing the proposed MCRS models. Section 5 presents experimental results and discussion. Finally, Sect. 6 concludes the study and outlines future research directions.

Background

Multi-criteria recommender systems

Compared to the conventional RSs, which typically rely on a single overall rating to model user preferences, MCRSs leverage multiple ratings across various criteria to provide a more nuanced understanding of user preferences. Specifically, the rating function in the MCRS is expressed as $R: U \times I \rightarrow R_0 \bigtimes R_1 \bigtimes R_2 \bigtimes \ldots \bigtimes R_c$, where R_0 is the overall rating and R_i (for $i = 1,\ldots,c$) represents the rating for each criterion i. Figure 1 illustrates an example of multi-criteria ratings in the movie domain, where each item has an overall rating and four sub-ratings based on the criteria of *story*, *cast*, *performance*, and *music*. In the figure, $\boldsymbol{u_1}$ is the target user for whom the system aims to predict the rating of movie $\boldsymbol{i_4}$. So, identifying the most similar users for $\boldsymbol{u_1}$ is the first step. Two key observations can be made from the data. First, while users $\boldsymbol{u_2}$ and $\boldsymbol{u_4}$ share the same overall ratings as $\boldsymbol{u_1}$, their ratings for individual criteria differ significantly. This suggests that their preferences or priorities for specific aspects vary considerably despite having similar overall evaluations. Second, users $\boldsymbol{u_3}$ and $\boldsymbol{u_5}$ exhibit preferences that align closely with $\boldsymbol{u_1}$ across all criteria, indicating high similarity in their evaluations. This contrast highlights the distinction between overall agreement and granular consistency in user preferences.

In a single-criterion system, the most similar users to the target user $\boldsymbol{u_1}$ would be $\boldsymbol{u_2}$ and $\boldsymbol{u_4}$, as they share similar overall ratings for movies $\boldsymbol{i_1}$, $\boldsymbol{i_2}$ and $\boldsymbol{i_3}$. However, their preferences are fundamentally different (For example, for movie $\boldsymbol{i_1}$, $\boldsymbol{u_1}$ dislikes the *story* and *cast* but highly values *performance* and *music*. In contrast, $\boldsymbol{u_2}$ and $\boldsymbol{u_4}$ love the *story* and *cast* but find *performance* and *music* unsatisfactory). This misalignment occurs because the single-criterion system cannot account for differences in how users evaluate items, leading to suboptimal recommendations. The system

Items / Users	$\boldsymbol{i_1}$	$\boldsymbol{i_2}$	$\boldsymbol{i_3}$	$\boldsymbol{i_4}$
$\boldsymbol{u_1}$	(**5**, 2, 2, 8, 8)	(**7**, 5, 5, 9, 9)	(**7**, 5, 5, 9, 9)	?
$\boldsymbol{u_2}$	(**5**, 8, 8, 2, 2)	(**7**, 9, 9, 5, 5)	(**7**, 9, 9, 5, 5)	(**9**, 8, 9, 10, 8)
$\boldsymbol{u_3}$	(**6**, 3, 3, 9, 9)	(**7**, 6, 6, 8, 8)	(**6**, 4, 4, 8, 8)	(**5**, 2, 2, 8, 8)
$\boldsymbol{u_4}$	(**5**, 7, 7. 3, 3)	(**7**, 8, 9, 4, 5)	(**7**, 9, 8, 5, 4)	(**9**, 7, 9, 9, 10)
$\boldsymbol{u_5}$	(**6**, 3, 3, 9, 9)	(**7**, 6, 6, 8, 8)	(**7**, 5, 5, 9, 9)	(**5**, 2, 3, 8, 7)

Fig. 1 An example of multi-criteria ratings

cannot distinguish between users with similar overall ratings but divergent preferences across specific criteria. In contrast, the MCRS can identify that $\boldsymbol{u_3}$ and $\boldsymbol{u_5}$ are better neighbours to $\boldsymbol{u_1}$ as their preferences align more closely across all criteria. The ability to capture fine-grained preferences results in more accurate and personalised recommendations.

In MCRS, criteria are the dimensions used to evaluate items. These criteria are classified into explicit and implicit [7]. Explicit criteria are predefined, structured, and easy to analyse but may not fully capture the diversity of user preferences due to their fixed nature. In contrast, the implicit criteria are dynamic, user-centric, and extracted from free-form text, offering a more flexible and nuanced understanding of user preferences. Table 1 provides a comparison between explicit/fixed and implicit criteria.

As shown in Table 1, implicit criteria provide a more flexible and user-centric approach to capturing user preferences, leading to potentially more accurate recommendations than relying solely on explicit criteria.

During the prediction phase, MCRSs incorporate these criteria to identify the most similar users (neighbours) for the target user. The prediction phase involves finding the most similar users for the target user and predicting ratings for unknown items using the identified users. Therefore, selecting relevant neighbours for the target user based on similarity computation will significantly impact prediction accuracy. MCRSs offer two methods for computing similarity [19]:

1. Conventional Similarity Measures: Calculate the similarity for each criterion individually (Eq. 1) and aggregate the results using a Cosine-based or Pearson correlation.

$$Sim(U_1, U_2) = \sum_{c=0}^{N} w_c sim_c(U_1, U_2) \tag{1}$$

where N denotes the number of criteria,w_c is the weight of each criterion to indicate its importance in the recommendation, and $sim_c(U_1, U_2)$ is the similarity between U_1 and U_2 on a single criterion indicated by c.

Table 1 Comparison between implicit and explicit criteria

Characteristic	Explicit Criteria	Implicit Criteria
Definition	Predefined and structured criteria that are provided by the system or experts	Extracted from user-generated reviews using NLP techniques
Flexibility	Fixed and limited to predefined criteria	Flexible and not constrained by a predefined set
Representation	Directly provided by users through structured ratings	Inferred from the free-form text of user reviews, capturing what users emphasise
Data Collection	Straightforward, as users rate predefined criteria	Requires advanced techniques (e.g., NLP, sentiment analysis) to extract criteria
User Preferences	It may not fully capture individual user priorities, as users are limited to predefined criteria	Better reflects user preferences, as users voluntarily mention what matters to them
Depth of Insights	Provides structured but potentially limited insights into user preferences	Offers richer, more nuanced insights into user opinions and priorities
Example	A hotel review form asking users to rate "Cleanliness" or "Service" on a 1–5 scale. While in a movie review system, explicit criteria could include "Story," "Cast," and "Music."	In a hotel review, implicit criteria might include "bed, staff, and breakfast," which are extracted from the review text rather than being part of a predefined rating system

2. Multi-dimensional Distance Metrics: Multi-dimensional distance measures like Manhattan distance (Eq. 2) or Euclidean distance (Eq. 3) are used to calculate similarity between users. The overall distance between the two users is the average distance between all shared items (I) (Eq. 4)

$$\text{Manhattan distance } (U_1, U_2) = \sum_{c=0}^{\mathrm{N}} |R_c(U_1, i) - R_c(U_2, i)| \tag{2}$$

$$\text{Euclidean distance } (U_1, U_2) = \sqrt{\sum_{c=0}^{\mathrm{N}} |R_c(U_1, i) - R_c(U_2, i)|} \tag{3}$$

$$dist(U_1, U_2) = \frac{1}{|\mathrm{I}(U_1, U_2)|} \sum_{i \in \mathrm{I}(U_1, U_2)} d(R(U_1, i), R(U_2, i)) \tag{4}$$

where N denotes the number of criteria, $R_c(U_1, i)$ is the rating of user U_1 for item i on criterion c, and $d(R(U_1, i), R(U_2, i))$ is the distance between users: U_1 and U_2 on item i.

Once neighbours are selected, MCRSs predict ratings for unrated items. The two most practical algorithms for rating prediction are the k-nearest neighbour (kNN) methods and the matrix factorisation-based methods [20, 21]. This study focuses on the kNN algorithm, which has evolved as one of the most popular fundamental algorithms for CF recommendation systems. There are various kNN approaches, including basic kNN, kNN that considers each user's z-score normalisation, and kNN that considers each user's mean ratings (kNNWithMeans). The kNNWithMeans approach, as given in (Eq. 5), was chosen for this study:

$$p(U_1, i) = \overline{r_{U_1}} + \frac{\sum_{x=1}^{n} sim(U_1, U_x).\left(r_{U_x,i} - \overline{r_{U_x}}\right)}{\sum_{x=1}^{n} sim(U_1, U_x)} \tag{5}$$

where $p(U_1, i)$ is the predicted rating for the user U_1 on item i; $\mathrm{sim}(U_1, U_x)$ is the similarity between users U_1 and U_x; $r_{U_x,i}$ is the rating given by the user U_x for item i; $\overline{r_{U_1}}$ is the average rating of the user U_1; and n is the number of nearest neighbours.

Review elements

Review elements are valuable information that can be extracted from user reviews to enhance recommendation systems. These elements can be accessed directly from the review page or extracted from unstructured textual reviews via opinion-mining techniques. Chen et al. [22] and Al-Ghuribi and Noah [7] identified seven review elements that are particularly useful for recommendations: *implicit ratings, frequent terms, review aspects/topics, contextual opinions, comparative words, review emoticons* and *review helpfulness*. Table 2 provides detailed descriptions of these elements, and Fig. 2 illustrates examples of these elements in a hotel review.

The review includes:

- **Context:** The sentence in blue highlights the context (e.g., "business").

Table 2 Description of review elements

Review Element	Description
Implicit rating	A score that is generated by aggregating the sentiment scores of opinion words in a review
Terms	Words that frequently appear in reviews
Aspects/Topics	Concepts (also called opinion targets) that describe specific features of an item, composed of related terms
Context	The situation or scenario in which a user provides their opinion about an item
Comparative Words	Words that are used to compare items based on specific features, such as "best" or "worst."
Emoticons	Symbolic icons (e.g., ,) that express a user's sentiment about an item
Helpfulness	The number of votes a review received indicates how useful readers found the review

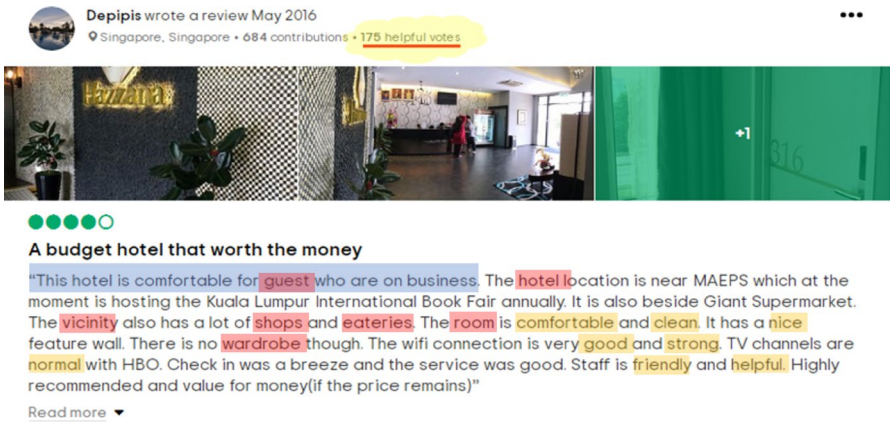


Fig. 2 Example of a hotel review from TripAdvisor

- **Review Terms:** Words in red represent frequently occurring terms (e.g., "guest," "hotel").
- **Sentiment Words:** Words in yellow indicate sentiment (e.g., "comfortable," "good").
- **Helpfulness:** The review received **175 helpful votes**, reflecting its usefulness to readers.

Related work

The field of recommender systems has evolved significantly over the past decade, with MCRS emerging as a prominent area of research. This section provides a comprehensive overview of existing works, highlighting their contributions, limitations, and how they relate to our proposed approach. We also discuss recent advancements in MCRS to contextualise our work within the current state of the art.

Traditional recommender systems

Traditional recommender systems primarily rely on single-criterion ratings, such as overall user preferences or binary feedback (e.g., likes/dislikes). Collaborative and content-based filtering are two widely adopted techniques in this domain [23]. CF methods leverage user-item interactions to predict preferences, while CB methods focus on item attributes to recommend similar items. However, these approaches often fail to capture

the reliable users' preferences, as they ignore the multi-dimensional nature of user feedback [24].

By considering the multi-criteria ratings, the RS can make more informed decisions about user similarities and preferences, leading to better-tailored recommendations.

Multi-criteria recommender systems

MCRS enhance traditional RSs by incorporating multiple criteria to provide more personalised and accurate recommendations. Adomavicius and Kwon [25] pioneered MCRS by introducing aggregation-based methods that combine multiple criteria into a single utility score. Subsequent research expanded on this seminal work, exploring techniques such as weighted aggregation to prioritise criteria based on user preferences [26], extension of matrix factorisation to include multi-criteria ratings [27], and hybrid approaches that leverage multi-criteria data for improved accuracy [28].

Recent studies, including those by Dridi et al. [6], Wasid and Ali [9], and Shambour [8] focused on developing MCRS using explicit criteria with datasets like TripAdvisor and Yahoo! Movies. For example, Dridi et al. [6] combined fixed criteria with contextual conditions to uncover user preferences, while Wasid and Ali [9] integrated multi-criteria ratings into CF using frequency count methods and particle swarm optimisation. Shambour [8] used deep learning to model hidden multi-criteria relationships among users. Similarly, other studies [29, 30] implemented deep learning-based MCRS that integrate explicit ratings and contextual information into a unified neural network, achieving superior performance on the TripAdvisor dataset.

Despite the effectiveness of explicit criteria, their reliance on fixed, predefined criteria may not fully capture user preferences, leading to issues like data sparsity and cold-start problems. To overcome these limitations, recent research has shifted towards exploiting implicit criteria derived from user reviews, providing a richer understanding of user preferences [22]. The review elements that are implicitly expressed in user reviews are referred to as implicit criteria. The two most common review elements are the *implicit ratings* and *aspects*.

Studies have explored *implicit ratings* from reviews, ranging from sentiment analysis using BERT for enhanced recommendations [12] to sentiment-aware CF approaches for Arabic reviews [11]. Other works have focused on extracting *implicit ratings* from reviews to improve CF performance on the Amazon dataset [31] or combining user ratings with *implicit ratings* to alleviate sparsity [32]. Pappas and Popescu [15] exploited *implicit ratings* to address the one-class CF problem and revealed a relationship between user ratings and their sentiments expressed in reviews.

Other studies focus on aspects, such as [13], which proposed a CF approach based on aspect clustering, which groups semantically similar aspects to reduce sparsity in the rating matrix, improving predictive accuracy on the Amazon movie dataset. Ray et al. [18] leveraged aspect-based sentiment analysis with BERT and Random Forest for hotel recommendations, categorising reviews effectively to provide personalised recommendations based on user queries.

Similarly, Liu et al. [33] used aspect-based sentiment analysis with dual attention mechanisms to improve rating prediction. The approach by Da'u et al. [34] derived ratings from aspects using deep learning and tensor factorisation, significantly enhancing

RS performance. Musto et al. [35] extracted aspects using a non-symmetric measure called Kullback–Leibler divergence and then fed the extracted aspects into the user-based and item-based CF algorithms. Akhtar et al. [36] classified users based on aspects and recommended items based on the users' similarities in each specified class.

Other review elements, such as *context* and *helpfulness*, have received less attention. Hu et al. [37] and Chen and Chen [38] integrated contextual information into their recommendation models, showing improved performance over traditional RS. Hasanzadeh et al. [39] and Park et al. [40] utilised review *helpfulness* to enhance recommendations. Mauro et al. [41] and Mitra and Jenamani [42] attempted to estimate the *helpfulness* value by presenting estimation models capable of improving RS performance. Similarly, *emoticons* have been used to model user emotions, as in Moshfeghi et al. [43] and Polignano et al. [44], who proposed emotion-aware recommendation models.

Overall, while MCRS show promise, most studies rely on explicit criteria, which often fail to capture nuanced user preferences, resulting in challenges like data sparsity and cold-start problems. In contrast, recent studies have demonstrated the effectiveness of leveraging implicit criteria in both the rating prediction and recommendation phases. However, existing studies typically focus on only one or two review elements, leaving a gap in understanding the combined impact of multiple elements. Integrating multiple review elements offers a more holistic view of user behaviour, potentially enhancing MCRS performance. This study addresses this gap by proposing models that combine multiple review elements, aiming to advance the state of the art in the rating prediction phase of MCRS and thus provide more robust and accurate recommendations.

Methodology

The primary objective of this study is to explore and evaluate various MCRS models by combining multiple review elements derived from user reviews. To achieve this, the experimental framework consists of four key phases: dataset preparation and enhancement, user profile modelling, MCRS model development, and rating prediction experiments.

Phase 1: dataset preparation and enhancement

Dataset selection

The Amazon Movie dataset was selected for its extensive scale, diversity, and rich review metadata, which are critical for extracting detailed user preferences and evaluating MCRS models [45]. This dataset has been extensively validated in prior RS research, providing a robust foundation for comparative analysis [21, 31]. The movie domain, notably prevalent in recent research [4, 21], is particularly suitable for this study due to its inherent multi-criteria nature (e.g., *story*, *cast*, and *performance ratings*), which aligns with the requirements of MCRS development. Figure 3 illustrates a sample record from the dataset, which includes nine attributes: *reviewerID, ProductID, reviewerName, helpful, reviewText, summary, overall, unixReviewTime,* and *reviewTime*.

Preprocessing

Unlike conventional RSs relying solely on an overall prediction rating, MCRSs incorporate multiple elements. This study focuses on the *helpful, aspect* and *implicit rating*

{"***reviewerID***" : "A1MPTJIR3LDGNZ",
"***ProductID***" : "780021088",
"***reviewerName***": "skipmccoy",
"***helpful***": [5,9],
"***reviewText***":' This is a charming version of the classic Dicken's tale. Henry Winkler makes a good showing as the ''Scrooge'' character. The casting is excellent and the music old but very relevant",
"***overall***": 5.0,
"***summary***": 'A smack-your-pimp-with-the-phone classic!",
"***unixReviewTime***": 967939200,
"***reviewTime***": " 09 3, 2000"}

Fig. 3 Example of Amazon dataset

elements, which are pivotal for modelling user preferences and item characteristics, thus making recommendations more relevant and personalised. The *aspect* and *implicit rating* are extracted and derived from the *reviewText* attribute of the dataset.

- **Aspect Extraction**

  The Semantically Enhanced Aspect Extraction (SEAE) approach is employed for its novelty and effectiveness relative to other baselines [46]. SEAE is a hybrid method that merges frequency-based and syntactic-relation-based approaches, further enhanced by a semantic similarity-based approach. The *Aspects* are represented as follows:

  $Aspects = \{(aspectName_i, aspectWeight_i, aspectRating_i, numberOfOccurrences_i) \mid i = 1..n$, where n is the number of aspects$\}$.

  The modified Term Frequency-Inverse Document Frequency (TF-IDF) weighting scheme developed by Zhu et al. [47] calculates the weight of each aspect (*aspectWeight*) as follows:and

$$modified - TFIDF = \frac{TF_{t,r}}{TF_r} \times \frac{\log(R)}{r_t} \times IF_t,$$

$$IF_t = \sqrt{\frac{1}{C}\sum_{i=1}^{C}\left(eq_{t,C_i} - \frac{1}{C}\right)^2} \tag{6}$$

  where C represents the total number of classes, and eq_{t,C_i} denotes the number of reviews within class C_i that contain term t. The dataset is segmented into five classes ($C=5$) based on the rating values assigned to each review. Additionally, the ratings of each aspect (*aspectRating*) are determined by the sentiment scores related to each aspect, calculated using the domain-specific lexicon proposed by Al-Ghuribi et al. [48].

- **Implicit Rating Calculation**

Two types of implicit ratings are being considered: implicit ratings based on aspect-sentiment word pairs and implicit ratings based on the sentiment scores of all the opinion words included in a review. The equations of both ratings are as follows:

$$ImplicitRating_{Aspect_i} = \sum_{j=1}^{N} aspectRating_j aspectWeight_j \tag{7}$$

where N is the number of aspects in review i, $aspectRating_j$ and $aspectWeight_j$ refer to the rating and weight for aspect j in review I,

$$ImplicitRating_{SWi} = \sum_{j=1}^{k} Score(SW_j) \tag{8}$$

where k is the number of sentiment words in review i and $Score(SW_j)$ is the sentiment score for the sentiment word j mentioned in the review, calculated using the lexicon proposed in [48].

Feature augmentation

The Amazon dataset is augmented with four new attributes: *Aspects*, aspect-based implicit ratings (*ImplicitRating_Aspect*), sentiment word-based implicit ratings (*ImplicitRating_SW*), and dominant aspect (*dominan_Aspect*), where *dominan_Aspect* refers to the aspect that has the highest rating.

To address sparsity and semantic redundancy in the *Aspects* attribute, aspects were clustered using Google's pre-trained Word2Vec model and K-means clustering. This process reduced dimensionality and grouped semantically related aspects (e.g., "plot" and "storyline").

Post-clustering, the dataset includes $13+K$ attributes, where K is the number of aspect clusters. Thus, the revised *Aspects* attribute is structured as follows:

$$Aspects = \left\{ \left(A_{i,j}, W_{A_{i,j}}, R_{A_{i,j}}, O_{A_{i,j}}, AR_i \right) | \left(i = 1..K, j = 1..n \right) \right\}$$

where $A_{i,j}$ is the name of the main aspect j in cluster i, and $W_{A_{i,j}}, R_{A_{i,j}}$, and$O_{A_{i,j}}$ are the weight, rating and occurrences of aspect $A_{i,j}$, respectively. AR_i refers to the average rating of all aspects in cluster i. The calculations of $W_{A_{i,j}}$ and $R_{A_{i,j}}$ follow the previously described method for *aspectWeight* and *aspectRating,* respectively. Figure 4 illustrates an example of the extended dataset structure.

Phase 2: user profile modelling

In this phase, a comprehensive user profile model is constructed using the augmented Amazon movie dataset, which includes 15 distinct features. These features are designed to capture a broad spectrum of user behaviours and preferences, and they are detailed in Table 3.

The UserProfiling algorithm depicted in Fig. 5 systematically assigns values to these features. It constructs comprehensive user profiles using the extended Amazon dataset and a preprocessed UserDetails list, which helps identify the unique number of users. This list is crucial for determining the size of each user's profile. The process

{"***reviewerID***‘: "A1MPTJIR3LDGNZ",
"***ProductID***‘: "780021088",
"***reviewerName***": "skipmccoy",
"***helpful***": [5,9],
"***reviewText***": "This is a charming version of the classic Dicken’s tale. Henry Winkler makes a good showing as the ‘‘Scrooge’’ character. The casting is excellent and the music old but very relevant",
"***overall***": 5.0,
"***summary***": 'A smack-your-pimp-with-the-phone classic!",
"***unixReviewTime***": 967939200,
"***reviewTime***": " 09 3, 2000"
"***Aspects: “version, tale, showing, casting, music”***
"***ImplicitRating_Aspect***”: “0.12022”
"*ImplicitRating_SW*": “1.1075”
"*dominant_Aspect*": “casting”}

Fig. 4 Example of the extended Amazon dataset

Table 3 Features of the user profile model

Features	Description
1. *User Identifier*	Unique ID assigned to each user
2. *User Trust Ratio*	Combines two metrics: - Trust Ratio: Classifies users as ‘Highly Trusted’, ‘Trusted’, or ‘Not Trusted’, based on the alignment between users’ explicit and implicit ratings - Helpful Value: Aggregates votes indicating how many users found the reviews valuable
3. *User Type*	Categorises users as ‘Optimist’ (positive reviews > negative), ‘Pessimist’ (negative > positive), or ‘Neutral’ (balanced ratings or majority neutral reviews)
4. *User Experience*	Reflects engagement level, calculated as: - Number of reviews posted - Number of movies watched
5. *Aspects*	List of aspects mentioned in reviews, each with metadata: - Aspect name - Aspect weight (TF-IDF score) - Aspect rating (sentiment score) - Occurrence frequency
6. *User seen movies*	List of *ProductID*s for movies the user has watched
7. *Explicit ratings*	List of all overall ratings provided by the user for movies
8. *Average explicit ratings*	Mean of *Explicit ratings*
9. *Implicit ratings from aspects*	Lists of all sentiment-derived ratings for movies based on aspect-sentiment pairs
10. *Average implicit ratings from aspects*	Mean of aspect-based implicit ratings
11. *Average of explicit and implicit ratings using aspects*	A combined average of explicit and aspect-based implicit ratings
12. *Implicit ratings from sentiment words*	Lists of all sentiment scores derived from opinion words in reviews
13. *Average implicit ratings from sentiment words*	Mean of sentiment word-based implicit ratings
14. *Average of explicit and implicit ratings using sentiment words*	A combined average of explicit and sentiment word-based implicit ratings
15. *Dominant aspect*	Identifies the most influential aspect for the user: - Highest-Rated Aspect - Most Frequently Mentioned Aspect

Algorithm UserProfiling **is**
Input: Extended Amazon Dataset, UsersDetails list
Output: *UsersProfile*.csv

1: Create Dictionary named as User_profile with Keys: Values equal to (123960 unique users: 15 attributes).
2: Prepare Dataset
A: Read the Amazon Dataset into DataFrame (DF) to ease its processing.
B: Sort DF based on reviewerID column to reduce the execution time.
3: Read *UserDetails* list
4: **For each** User (Ui) in *UserDetails* list **Do**
A: Assign Zero value for all the variables Seenmovies, Helpful, Trust, aspects, User Type variables, rating variables
B: **For each** row (j) in DF related to Ui **Do**
a: Determine the seen movies for Ui: Seenmovies.append (DF.ProductID[Ui][j])
b: Sum the Helpfulness value: Helpful+ ← DF.helpful[Ui][j]
c: Check Trust level:
IF ((DF.overall[i] in [4,5]) **AND** ((DF.ImplicitRating_Aspect[i]>0) **OR** (DF.ImplicitTotalSentimentWord[i]>0)))**OR**(((DF.overall[i] in [1,2]) **AND** ((DF.ImplicitRating_Aspect[i]<0) **OR** (DF.ImplicitTotalSentimentWord[i]<0))))
Trust+ ← 1
d: Identify User Type:
IF (DF.overall[i] in [4,5]) Optimistic+ ←1
ELIF (DF.overall[i] = [3]) Neutral+ ← 1
ELIF (DF.overall[i] in [1,2]) Pessimistic+ ← 1
e: Identify the aspects for Ui from column DF.Aspects[Ui], their ratings and count the occurrences of each aspect
aspects (name, rating, occurrence)
f: Append ratings into lists:
Explicit_Rating.append (DF.overall[Ui][j])
Implicit_Rating_Aspect.append (DF.ImplicitRating_Aspect[Ui][j])
Implicit_Rating_Sentiment_Words. append (DF.ImplicitTotalSentimentWord[Ui][j])
C: Insert the user (Ui) information into the User_profile with Key DF.reviewerID[Ui] with the following values:
a: Seenmovies.
b: Add user (Ui) Trust based on the Trust value compared to the number of seen movies.
c: Add user (Ui) Type based on the max value among the Optimistic, Neutral, and Pessimistic values.
d: Add user (Ui) Experience based on the number of seen movies and reviews posted by Ui.
e: Explicit_Rating, Implicit_Rating_Aspect, Implicit_Rating_Sentiment_Words.
f: Average ratings: explicit, implicit using aspects and using sentiment words.
g: Average explicit rating with each implicit rating (aspect and sentiment word) separately.
h: aspects, dominant aspect with max aspect rating from aspects.
5: **Save** User_profile dictionary to UsersProfile.csv

Fig. 5 User profiling algorithm

begins by initialising a dictionary (User_profile) to store user entries, each with 15 attributes. The dataset is then loaded into a sorted DataFrame to optimise processing efficiency. For each user, the algorithm methodically processes their review data to perform the following tasks:

1. **Aggregate behavioural metrics**: Track seen movies (*ProductID*), helpfulness votes, and user experience (number of reviews/movies).
2. **Determine trustworthiness**: Classify users as trusted if their explicit ratings (e.g., 4–5 stars) align with positive implicit sentiment scores (aspect/sentiment-based ratings) or untrusted if low ratings align with negative sentiment.
3. **Categorise user type**: Label users as *Optimistic* (mostly high ratings), *Pessimistic* (mostly low ratings), or *Neutral* (balanced/mixed ratings).
4. **Extract aspects and ratings**: Collect aspect names, weights, sentiment scores, and occurrence frequencies from reviews.
5. **Compute averages**: Derive mean values for explicit ratings, implicit aspect-based ratings, and implicit sentiment word-based ratings.
6. **Identify dominant aspects**: Highlight the aspect with the highest rating or frequency for each user.

Once all necessary data is collected and calculated, it is compiled into the "User_profile" dictionary. This dictionary is then exported to a file named UsersProfile, which is ready for use in downstream multi-criteria recommendation tasks. This pipeline ensures robust, granular user modelling by integrating explicit ratings, implicit sentiment-derived scores, and behavioural insights—such as trustworthiness classifications and dominant aspect identification. The resultant profiles empower the recommendation system to generate highly personalised predictions grounded in understanding user preferences and behavioural patterns.

Phase 3: MCRS models development

This phase develops diverse recommendation models by integrating multiple review elements into the *k*NN algorithm. The *k*NN algorithm is strategically chosen for its ability to:

- compute multi-dimensional similarities across diverse review elements;
- prioritise localised patterns in large-scale datasets like Amazon, mitigating noise from sparse ratings; and
- adapt flexibly to hybrid criteria combinations through customisable distance metrics.

We consider four user-centric types, each leveraging distinct review elements.

a. **Trusted Users**

Three different MCRS models are proposed for this type of user using the *'User Trust Ratio'* and '*Aspects'* attributes from the proposed user profile. As indicated in Table 3, the *'User Trust Ratio'* attribute has two values: trust ratio and helpfulness value. The trust ratio shows the degree of similarity of the sentiment orientation between the user's implicit ratings and overall ratings (i.e. explicit ratings) for the same item. Ratings are considered positive if ≥ 4 and negative if ≤ 2. Ratings $= 3$ are considered neutral. Based on how many seen movies have similar implicit and explicit sentiment orientation for a particular user (i.e., both ratings are positive or negative), the user is categorised as *'Highly Trusted'*, *'Trusted'*, or *'Not Trusted'*, based on the following agreement score (Eq. 9) and (Eq. 10), where a_i and e_i refer to the sentiment orientation of the implicit ratings and the overall ratings of the i-th movie, respectively (i.e. positive or negative), s_i be the agreement score for the i-th movie, and N is the total number of movies rated by the user.

$$TotalAgreementScore(AS) = \sum_{i=1}^{N} s_i \begin{cases} s_i = 1 if a_i = e_i \\ s_i = 0 if a_i \neq e_i \end{cases} \tag{9}$$

$$UserTrust = \begin{cases} Highlytrusteduser, if \frac{AS}{N} = 1 \\ Trusteduser, if 0.5 \leq \frac{AS}{N} < 1 \\ Nottrusteduser, if \frac{AS}{N} < 0.5 \end{cases} \tag{10}$$

The helpfulness value, on the other hand, is the sum of all the helpfulness values (i.e. the attribute *helpful* in the Amazon dataset) assigned to the user.

A different combination of the attributes is suggested to be integrated as multiple criteria in the MCRS for rating prediction in each proposed model. The proposed MCRS models for the Trusted Users are as follows.

i. **Trusted Multi-Criteria Similarity (TMCS)**–Neighbours in this model are the *Trusted* and *Highly Trusted* users with high *helpful* value. The similarity is in terms of average ratings of aspects of co-rated items, as shown in (Eq. 11), where $I(u_x, u_y)$ refers to the set of items rated by both u_t and u_x. $AR^{u_x}_{m_i,c_j}$ and $AR^{u_y}_{m_i,c_j}$ refer to the average ratings of clustered aspect c_j for users u_x and u_y respectively on item m_i.

$$dist(u_x, u_y) = \frac{1}{|I(u_x, u_y)|} \sqrt{\sum_{i=1..t} \sum_{j=1..n} \left| AR^{u_x}_{m_i,c_j} - AR^{u_y}_{m_i,c_j} \right|} \tag{11}$$

As a result, the selected neighbours are k users who are trusted and supportive, with a high similarity to the target user. The similarity relates to the same preferences for the aspects and ratings of the shared items.

ii. **Trusted Single-Criterion Similarity (TSCS)** – The similarity in this model is calculated based on a single criterion representing a user's overall rating of a consumed item. $R^{u_x}_{m_i,o}$ and $R^{u_y}_{m_i,o}$, respectively refer to the overall rating of item m_i given by user u_x and u_y.

$$dis(u_x, u_y) = \frac{1}{|I(u_x, u_y)|} \sqrt{\sum_{i=1..t} \left| R^{u_x}_{m_i,o} - R^{u_y}_{m_i,o} \right|} \tag{12}$$

iii. **Trust-based Multi-Criteria Similarity with Average Value (TMCSAV)**—The similarity between users for this model is similar to the TMCS model. However, only the k *Highly Trusted* or *Trusted* users with the highest Trust Average (TAV) values are selected as neighbours for the target user U_t. TAV is defined as (Eq. 13).

$$TAV(u_t, u_x) = \frac{1}{2}\left(Helpful_x + dist(u_t, u_x)\right) \tag{13}$$

This model limits neighbours to only those with a high average trust value and multi-criteria similarity with a target user on different aspects.

b. **Experienced Users**

Experienced users are defined by high engagement (review count/movie views). This type of model mainly focuses on the *User Experience* and *overall* attributes. The *User Experience* attribute contains two values: the number of reviews posted by the user and the number of movies the user has viewed. The similarity between users is similar to the TMCS model. Two models are proposed for this type of user:

i. **Experienced Multi-Criteria Similarity (EMCS)**—Neighbours are k experienced users with the highest number of reviews posted. The similarity between users is similar to the TMCS model based on (Eq. 11).

ii. **Experience-based Multi-Criteria Similarity with Average Value (EMCSAV)**—The similarity for this model is similar to the EMCS model. However, only the k experienced users with the highest Experience Average (EAV) values are selected as neighbours. EAV is defined as follows: where $Views_x$ refers to the number of movies the user u_x has viewed.

$$EAV(u_t, u_x) = \frac{1}{2}(Views_x + dist(u_t, u_x)) \tag{14}$$

c. **Optimistic Users**

Optimistic users are defined based on outnumbering positive reviews over negative reviews.

This type of model mainly focused on the *User Type* feature. A user is considered optimistic if the number of positive reviews is larger than the number of negative reviews. Two multi-criteria models are proposed based on the optimistic user model.

i. **Optimistic Multi-Criteria Similarity (OMCS)**—This model considers neighbours to be top k optimistic users with high similarity. The similarity metric is similar to the TMCS model (Eq. 11).

ii. **Optimistic-Trusted Hybrid (OTMCSAV)**—This type of model refers to neighbours that are optimistic and trusted. Thus, only the k optimistic users with the highest TAV values are selected as neighbours for this model. The similarity metric is similar to the TMCS model (Eq. 11).

d. **Dominant Aspect-Based Users**

Users sharing the same influential aspect (most frequently mentioned/highly rated aspect) are the target of this type. This model is proposed using the '*Dominant aspect*' and '*overall*' attributes from the proposed user profile. Only a single multi-criteria model is being proposed based on the dominant aspect: the **Dominant Aspect Similarity (DSCS) model**, where neighbours are the top k users sharing the target user's '*Dominant aspect*' with high single-criterion similarity. The similarity metric is similar to the TSCS model (Eq. 12).

Table 4 summarises all the proposed MCRS models based on the four user-centric types previously described.

Rating prediction experiments

These experiments aim to evaluate the efficacy of each proposed model in terms of prediction accuracy against three baseline models:

- **Conventional Single-Criterion Recommender System (SRS)**: This model uses overall ratings as its sole criterion for recommendations.
- **Fixed Aspect-Based Recommender System (FixMCRS)**: Developed by Hernández-Rubio et al. [49], this system utilises predefined aspects to enhance recommendation accuracy.

Table 4 Summary of user-centric models

User-centric types	Multi-criteria models	Similarity metric	Neighbours
Trusted users	Trusted Multi-Criteria Similarity (TMCS)	Multi-dimensional distance across clustered aspects of co-rated items (Eq. 11)	Top *k Trusted/highly Trusted* users with high helpfulness and similarity
	Trusted Single-Criterion Similarity (TSCS)	Euclidean distance based on overall ratings (Eq. 12)	Top *k Trusted/Highly Trusted* users with high helpfulness and single-criterion similarity
	Trust-based Multi-Criteria Similarity with Average Value (TMCSAV)	Same as TMCS (Eq. 11)	Top *k* users with the highest Trust Average (*TAV*) values for the target user
Experienced Users	Experienced Multi-Criteria Similarity (EMCS)	Same as TMCS (Eq. 11)	Top *k* experienced users with high similarity
	Experience-based Multi-Criteria Similarity with Average Value (EMCSAV)	Same as TMCS (Eq. 11)	Top *k* users with highest Experience Average (*EAV*) values
Optimistic Users	Optimistic Multi-Criteria Similarity (OMCS)	Same as TMCS (Eq. 11)	Top *k* optimistic users with high similarity
	Optimistic-Trusted Hybrid (OTMCSAV)	Same as TMCS (Eq. 11)	Top *k* optimistic users with highest *TAV*
Dominant Aspect-Based Users	Dominant Aspect Similarity (DSCS)	Same as TSCS (Eq. 12)	Top *k* users share the target user's '*Dominant aspect*' with high single-criterion similarity

Table 5 Amazon Dataset Overview

Items	Total
Total number of reviews	1,697,533
Empty reviews	62
Total number of processed reviews	1,697,471
Number of unique users	123,960
Number of unique movies	50,052

- **Multi-Criteria Recommender System (CapMCRS)**: Proposed by Musto et al. [35], this model extracts and utilises dynamic aspects from user reviews using the proposed method by Caputo et al. [50] to tailor recommendations.

All models were implemented on a Windows 10 operating system, equipped with an Intel Core i5 processor and 8.00 GB RAM. The experiments utilised Python programming language (version 3.7.1, 64-bit) with development environments including Spyder 4.1 and Google Colab.

The evaluation leveraged the Amazon dataset, a widely recognised benchmark in RSs research, comprising approximately 1.7 million reviews across 123,960 unique users and 50,052 unique movies (Table 5).

The extended Amazon dataset with 13 attributes was utilised to build the users' profiles. The user profile data includes 123,960 records that give more fine-grained information about the users' preferences. As previously explained, these profiles

incorporated explicit ratings, implicit criteria, and behavioural metrics such as trust ratios and user experience. The key steps include:

1. Aspect Extraction: Using the SEAE method, 49 main aspects and 481 core terms were identified from reviews.
2. Aspect Clustering: Aspects were grouped into eight semantically coherent clusters using Google's Word2vec embeddings and K-means clustering (optimal $K=8$ determined via prior experiments [13]).

Models implementation

To mitigate data sparsity and ensure reliability in all experiments, the Amazon dataset was filtered to retain only movies with ≥ 20 ratings and users with ≥ 20 reviews, yielding a refined subset of 13,214 users, 17,022 movies, and 650,145 reviews.

For the proposed models, 49 main aspects and 481 core terms were extracted from the dataset using SEAE and clustered into eight clusters, as previously mentioned. Consequently, the proposed models utilised the extended Amazon dataset containing 21 ($13+K$ number of clusters) attributes and user profiles. On the other hand, for the baseline models, the FixMCRS model incorporated 23 main aspects with 271 core terms from Hernández-Rubio et al. [49], while the CapMCRS model utilised 223 aspects extracted via Caputo's method [50].

Both the FixMCRS and CapMCRS models modified the Amazon dataset by integrating relevant aspects into the '*Aspects*' attribute. Aspects are annotated with sentiment scores using the SentiWordNet lexicon, ensuring consistency across models. Aspects are then grouped into clusters reflecting their semantic similarities, a process facilitated by the use of Google's Word2vec and K-means clustering. This method is applied to the dataset, which is subsequently extended with eight additional attribute clusters. Consequently, the performances of all proposed and baseline models are evaluated using the enhanced Amazon dataset, and in each model, the required criteria are integrated for rating prediction. For the SRS model, no additional dataset modifications are necessary as it relies solely on overall ratings without aspect integration.

The kNNWithMeans algorithm (Eq. 5) was employed for rating prediction. The baseline and proposed models were tested with various neighbourhood sizes ($k=5$, 10, 20, 30, 40, and 50). Our findings revealed that a neighbourhood size of $k=50$ achieved the lowest error rates, indicating superior performance in terms of rating prediction. This optimal setting was selected for the final implementation.

Models evaluation

The proposed and baseline models employed the kNNwithMeans prediction algorithm for rating prediction, leveraging user similarity computed from different criteria. We integrated prediction and classification metrics to comprehensively assess the proposed models.

The prediction metrics metric are based on three error metrics: Mean Absolute Error (MAE), Mean Square Error (MSE), and Root Mean Squared Error (RMSE).

The RMSE value is the root mean square error, with a set of n actual ratings p and their predicted ratings $\overset{'}{p}$. The equation is as follows.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(p_i - \overset{'}{p}_i\right)^2} \tag{15}$$

The MAE and MSE are simply the mean absolute error and the average of the squares of the errors, respectively, where the equations are as follows.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|p_i - \overset{'}{p}_i\right| \tag{16}$$

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(p_i - \overset{'}{p}_i\right)^2 \tag{17}$$

For the evaluation of classification metrics, the focus is on measuring how effectively the recommender system prioritises items that users are likely to find relevant. In this context, relevance is determined using a binary threshold: items with ratings ≥ 4 (on the Amazon dataset's 1–5 scale) are classified as relevant, while items with ratings < 4 are considered irrelevant. Precision, recall, and F-score metrics were employed to assess performance. The equations for these metrics are as follows, respectively.

$$Precision(P) = \frac{N_{rs}}{N_s} \tag{18}$$

$$Recall(R) = \frac{N_{rs}}{N_r} \tag{19}$$

$$F1 - Score = \frac{2PR}{P + R} \tag{20}$$

Here, N_{rs} denotes relevant items correctly recommended, N_s is the total items recommended, and N_r is the total relevant items in the dataset. These metrics reflect real-world scenarios where users engage primarily with top-ranked recommendations. High precision ensures recommendations are trustworthy, while high recall ensures comprehensive coverage of relevant items.

Results and discussion

Experimental results

The proposed MCRS models were evaluated using five-fold cross-validation (80:20 train-test split) on the extended Amazon dataset. Table 6 summarises the results of the prediction error metrics of all the proposed multi-criteria models and the baselines. It shows that this study's proposed models outperform the three baselines, with the TMC-SAV giving the best prediction accuracy in terms of all the measured error metrics: MAE, MSE, and RMSE.

Table 6 Prediction accuracy of the proposed and baseline models

Method	MAE	MSE	RMSE
TMCSAV	**0.7473**	**0.9932**	**0.9966**
TMCS	0.7483	0.9974	0.9987
TSCS	0.7484	0.9982	0.9991
EMCS	0.7478	0.9974	0.9987
EMCSAV	0.7474	0.9938	0.9969
OMCS	0.7506	0.9993	0.9996
OTMCSAV	0.7502	0.9991	0.9996
DSCS	0.7486	0.9950	0.9975
SRS	0.8142	1.1576	1.0759
FixMCRS	0.8235	1.1690	1.0812
CapMCRS	0.8195	1.1640	1.0789

Table 7 demonstrates the performance improvement of the TMSAV model compared to the baseline models. Following this, Table 8 presents the classification performance results in terms of precision, recall, and F1-score, where items with ratings ≥ 4 are treated as relevant. To streamline the presentation and facilitate easier comparison, only the best-performing model is shown alongside the baseline results. This approach ensures a clear and concise evaluation of the models' effectiveness.

Key observations and findings

The following are the key findings and observations of the above results.

- **Trust-based multi-criteria similarity with average value model**

  The TMCSAV model demonstrated superior performance by achieving the lowest prediction errors, with MAE, MSE and RMSE at 0.7473, 0.9932, and 0.9966, respectively, as well as the highest classification performance, reflected in an F1-score of 0.65. These results indicate that the TMCSAV model outperformed all baseline

Table 7 TMCSAV improvement over baselines

Baseline	MAE Improvement (%)	MSE Improvement (%)	RMSE Improvement (%)
SRS	8.95	16.55	7.96
FixMCRS	10.20	17.70	8.49
CapMCRS	9.66	17.20	8.26

Table 8 Classification performance

Model	Precision	Recall	F1-Score
TMCSAV	0.68	0.62	0.65
SRS	0.45	0.42	0.43
FixMCRS	0.52	0.48	0.50
CapMCRS	0.58	0.55	0.56

models and other proposed approaches. The success of the TMCSAV model can be attributed to two key factors:

- **Trustworthy neighbours**: The model prioritises users with high trust scores, which are determined by the alignment of their explicit and implicit ratings, along with helpfulness metrics. This approach minimises the influence of unreliable reviewers and reduces noise in the data, leading to more accurate predictions.
- **Aspect-driven similarity**: The model enables fine-grained preference matching by leveraging 49 semantically clustered aspects extracted using SEAE and Word2vec. This aspect-driven approach allows for a more detailed and better understanding of user preferences, enhancing the model's ability to generate relevant recommendations.

- **Complementary Role of Experience**
  The experience-based model, represented by the EMCSAV model, secured the second-highest performance, underscoring the complementary role of user experience in refining recommendations alongside trust. While the EMCSAV model achieved marginally higher prediction errors compared to the top-performing TMCSAV model, its strong performance highlights the importance of incorporating user experience as a key factor in recommendation systems. This result also emphasises the critical role of trust in addressing challenges posed by sparse data, as trust metrics help mitigate the impact of data scarcity by prioritising reliable and consistent user interactions. Together, these insights demonstrate that combining user experience with trust-based mechanisms can significantly enhance the accuracy and reliability of recommendation systems.
- **Impact of User Experience and Trust**
  Models that integrate user experience (EMCS, EMCSAV) and trust (TMCS, TMCSAV) consistently achieved better performance compared to models relying on a single criterion, such as DSCS, or those based on optimistic user assumptions, like OMCS. For example, the EMCSAV model, which combines user experience with aspect similarity, achieved a low MAE of 0.7474, securing the second-highest rank. This demonstrates that incorporating both user experience and fine-grained aspect-driven similarity enhances recommendation accuracy. In contrast, the OTMCSAV model, which relies on optimistic assumptions, underperformed with an MAE of 0.7502 compared to the TMCSAV model. This highlights the limitations of optimistic assumptions in improving recommendation quality.
- **Dominant Aspect Efficacy**
  The DSCS model, which integrates the dominant aspect (highest-rated/most frequent aspect per user), reduced MAE by 8.06%, MSE by 14.05%, and RMSE by 7.29%, compared to SRS. This validates that even a single implicit criterion (dominant aspect) significantly improves accuracy over explicit-only models and underscores the necessity of multi-criteria integration.
- **Implicit versus Fixed Criteria**

The comparison between implicit and fixed criteria highlights significant differences in performance among the models. FixMCRS, which relies on fixed aspects, performed the worst among the baseline models, achieving an MAE of 0.8235 and an F1-score of 0.50. This poor performance can be attributed to its rigid criteria, which fail to adapt to the unique preferences of individual users, limiting its ability to provide personalised recommendations. On the other hand, CapMCRS, which utilises syntax-based aspects, showed improvement over FixMCRS with an F1-score of 0.56. However, it still underperformed compared to the TMCSAV model. This is primarily due to its inefficient aspect extraction method, which results in aspects that inadequately represent user preferences. These findings underscore the importance of dynamic and user-centric approaches, such as those employed by TMCSAV, in effectively capturing and leveraging user preferences for more accurate and personalised recommendations.

Theoretical and practical implications

The TMCSAV model represents a significant advancement in recommendation systems theory, addressing critical gaps in existing multi-criteria frameworks. Theoretically, it introduces three key innovations. First, its aspect clustering and semantic enrichment approach organises fine-grained aspects, such as grouping "battery life" under "device performance" into a hierarchical semantic framework. It extends NLP theory, enabling more interpretable and context-aware recommendations. Second, the model incorporates a trust-aware recommendation paradigm, leveraging dynamic trust metrics like reviewer helpfulness to align with social theories of trust as a fluid and evolving construct. Third, its multi-view learning architecture, which integrates text, ratings, and trust data, effectively tackles data sparsity, particularly in cold-start scenarios, reducing dependency on dense rating matrices. These theoretical contributions pave the way for more robust and adaptable recommendation systems.

Practically, the TMCSAV model and related MCRS frameworks offer substantial value across industries reliant on personalised recommendations. In e-commerce, for instance, the model can analyse user preferences for specific product features, such as "battery life" or "durability," while incorporating trust metrics to prioritise credible reviews. This enhances recommendation accuracy, reduces decision fatigue, and boosts customer satisfaction.

Streaming services benefit from the model's ability to refine recommendations based on subtle preferences, such as "character development" or "soundtracks," leading to higher user engagement and retention. Social networks can leverage the model to recommend relevant posts or groups by focusing on dominant aspects like "travel tips" or "tech news" while filtering out low-quality content, thereby fostering community engagement. In the hospitality and travel sector, the model's ability to match situational needs, such as prioritising "cleanliness" and "location" for a business trip versus a family vacation, results in higher booking rates and positive user experiences. Collectively, these applications demonstrate the TMCSAV model's potential to transform industries by delivering highly personalised and trustworthy recommendations.

Limitations

The study acknowledges several limitations that impact the performance and generalizability of the proposed models. First, the marginal performance differences between models, such as the small gap between TMCSAV and EMCSAV, can be attributed to three key factors. All models share a common foundation, utilising kNNWithMeans and multi-criteria profiles, which inherently limits the extent of variation in their results. Additionally, the Amazon dataset's large scale, comprising approximately 650,000 reviews, provides abundant signals that lead to performance convergence across models. Furthermore, the overlap between trust and experience metrics, such as the tendency for trusted users to also be experienced, reduces the distinctiveness of these factors, contributing to the narrow performance gaps.

Second, the suboptimal classification metrics, with precision and recall ranging between 0.42 and 0.68, are influenced by several challenges. Sparse labelling in public datasets, which lack fine-grained aspect annotations, limits the models' ability to capture nuanced user preferences. Class imbalance, resulting from binary thresholds (e.g., considering ratings ≥ 4 as positive), skews sentiment distributions, with positive reviews often dominating the dataset. Lastly, a mismatch between existing datasets and the requirements of dynamic multi-criteria recommendation systems (MCRS) poses a significant hurdle, as these datasets were originally designed for fixed-criteria systems rather than the more flexible and adaptive approaches proposed in this study. These limitations highlight areas for future improvement, such as developing richer datasets and refining model architectures to better address these challenges.

Conclusion and future works

This study addresses critical limitations in conventional recommender systems by proposing Multi-Criteria Recommender System (MCRS) models that integrate diverse review elements to refine user preference modelling. Evaluated against three baselines on the Amazon dataset, the proposed models demonstrate superior performance in both rating prediction (MAE/RMSE) and relevance identification (F1-score). These results validate the hypothesis that multi-criteria frameworks leveraging review-driven signals significantly outperform single-criterion or static approaches. The TMCSAV model, which synthesises trust-based neighbour selection, semantic aspect clustering, and user experience, emerges as the most robust solution, effectively mitigating data sparsity and enhancing recommendation reliability. Its success highlights the value of aligning explicit ratings with implicit sentiment and contextualising user preferences through semantically grouped features. While the study underscores the potential of review-driven MCRS, limitations in public datasets—such as sparse aspect annotations and imbalanced sentiment distributions—constrain generalizability of the proposed models. Future research will focus on utilising curated datasets with fine-grained annotations to better capture multi-criteria dynamics. Additionally, hybrid architectures combining TMCSAV's interpretable framework with deep learning techniques (e.g., transformer-based aspect extraction) could amplify robustness, while cross-domain applications (e.g., healthcare platforms) may reveal novel challenges in trust calibration and aspect interpretability. We also plan to expand our investigation by exploring a wider variety

of combinations of elements found in user profiles and examining their impact on the accuracy of MCRS performance. By bridging theoretical innovation with practical scalability, this work establishes a foundation for adaptive, user-centric systems capable of evolving with dynamic preferences, offering transformative potential for industries reliant on personalised recommendations, from e-commerce to streaming services.

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Author contributions
SMAG took on the main role of performing the literature, designing, performing experiments, and analysing data. SMAG and SAMN wrote the paper. SAMN and ST supervised the research. SAMN secured the funding for the study. MAM and NIYS reviewed the manuscript language and helped edit the manuscript. All authors read and approved the final manuscript.

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Data availability
The Amazon Dataset collected by Julian McAuley and is available at UC San Diego: https://cseweb.ucsd.edu/~jmcauley/datasets.html#amazon_reviews

Declarations

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Not applicable.

Consent for publication
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